About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE 2012 MATHEMATICS

A. Formulae

1.1.1 For real a, the principal values satisfy

$$\tan^{-1} a + \cot^{-1} a = \frac{\pi}{2} \tag{1.1.1.1}$$

Solution: Let $\theta = \tan^{-1} a$, so that $\tan\theta = a$ and $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $\frac{\pi}{2} - \theta \in (0, \pi)$, which is the range of $\cot^{-1}$, and

$$\begin{aligned} \cot\left(\frac{\pi}{2} - \theta\right) &= \tan\theta \\ &= a \\ \implies \cot^{-1} a &= \frac{\pi}{2} - \theta \end{aligned} \tag{1.1.1.2}$$

Adding $\theta = \tan^{-1} a$ to both sides gives (1.1.1.1).

1.1.2 For a square matrix A,

$$(I + A)^2 = I + 2A + A^2 \tag{1.1.2.1}$$

1.1.3 A system of n linear equations in n unknowns is written as

$$Ax = b \tag{1.1.3.1}$$

It has a unique solution $x = A^{-1}b$ if and only if $\left|A\right| \neq 0$. The solution is obtained by reducing the augmented matrix $\begin{pmatrix} A & b \end{pmatrix}$ to an upper triangular form using row operations (Gaussian elimination) followed by back substitution.

Solution: If $\left|A\right| \neq 0$, then A^{-1} exists. Multiplying $Ax = b$ on the left by A^{-1},

$$A^{-1}Ax = A^{-1}b \implies x = A^{-1}b \tag{1.1.3.2}$$

so every solution equals $A^{-1}b$, and this vector does satisfy the system. Hence the solution exists and is unique. Conversely, if $\left|A\right| = 0$, there is a vector $z \neq 0$ with $Az = 0$. Then whenever x_0 is a solution, so is $x_0 + tz$ for every real t, so the solution is not unique.

Each row operation multiplies both sides by an invertible matrix E, so $(EA)\,x = Eb$ has the same solutions as $Ax = b$. Reducing to $Ux = c$ with U upper triangular, $\left|U\right| = \prod_{i=1}^{n} u_{ii}$ is a non-zero multiple of $\left|A\right|$, so every $u_{ii} \neq 0$. The last row gives $x_n = \frac{c_n}{u_{nn}}$, and back substitution gives

$$x_i = \frac{1}{u_{ii}}\left(c_i - \sum_{k=i+1}^{n} u_{ik}x_k\right), \quad i = n-1, \ldots, 1 \tag{1.1.3.3}$$

The system $Ax = b$ is consistent if and only if

$$\operatorname{rank} A = \operatorname{rank}\begin{pmatrix} A & b \end{pmatrix} \tag{1.1.3.4}$$

and then the solution is unique if and only if this common rank equals n, the number of unknowns.

Solution: Write $A = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix}$ in terms of its columns, so that $Ax = x_1 a_1 + \cdots + x_n a_n$. The system has a solution if and only if b is a linear combination of the columns of A, that is, if and only if appending b does not increase the number of independent columns, which is (1.1.3.4). If the common rank is n, the columns are independent and the coefficients $x_1, \ldots, x_n$ are unique. If it is less than n, there is $z \neq 0$ with $Az = 0$, and the solution is not unique.

1.1.4 Determinants.

a) Adding a multiple of one row to another row does not change the determinant.

$$\left|EM\right| = \left|M\right|, \quad E = I + \lambda e_i e_j^\top, \; i \neq j \tag{1.1.4.1}$$

Solution: Since $e_j^\top M$ is row j of M, the product $EM = M + \lambda e_i \left(e_j^\top M\right)$ is M with λ times row j added to row i. As $i \neq j$, E is triangular (upper if $i < j$, lower if $i > j$) with unit diagonal, so from (1.1.4.3) $\left|E\right| = 1$ and

$$\begin{aligned} \left|EM\right| &= \left|E\right| \left|M\right| \\ &= \left|M\right| \end{aligned} \tag{1.1.4.2}$$

b) The determinant of a triangular matrix is the product of its diagonal entries.

$$\left|U\right| = \prod_{i=1}^{n} u_{ii} \tag{1.1.4.3}$$

Solution: Let U be upper triangular, so $u_{ij} = 0$ for $i > j$. Only u_{11} is non-zero in the first column, so expanding along it, $\left|U\right| = u_{11} \left|U_1\right|$, where U_1 is the upper triangular matrix of order $n-1$ obtained by deleting row 1 and column 1. Repeating this n times,

$$\left|U\right| = u_{11} u_{22} \cdots u_{nn} \tag{1.1.4.4}$$

For a lower triangular L, the matrix $L^\top$ is upper triangular and $\left|L\right| = \left|L^\top\right|$, so the same result holds.

c) If a row of M is a multiple of another row, then

$$\left|M\right| = 0 \tag{1.1.4.5}$$

Solution: Let row i of M be k times row j, with $i \neq j$. Applying (1.1.4.1) with $\lambda = -k$, row i of EM becomes $k\,\text{row}_j - k\,\text{row}_j = 0$, and

$$\begin{aligned} \left|M\right| &= \left|EM\right| \\ &= 0 \end{aligned} \tag{1.1.4.6}$$

because a matrix with a zero row has determinant 0 (expand along that row).

1.1.5 For differentiable f,

$$\int e^x \left(f(x) + f'(x)\right) dx = e^x f(x) + c \tag{1.1.5.1}$$

Solution: By the product rule,

$$\begin{aligned}\frac{d}{dx}\left(e^x f(x)\right) &= e^x f(x) + e^x f'(x) \\ &= e^x \left(f(x) + f'(x)\right)\end{aligned} \tag{1.1.5.2}$$

Integrating both sides with respect to x, the left side gives $e^x f(x)$ and the right side is the integral in (1.1.5.1), up to the constant c.

1.1.6 The area of the parallelogram with adjacent sides a and b is

$$\|a \times b\| \tag{1.1.6.1}$$

where, for $a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$,

$$a \times b = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix} \tag{1.1.6.2}$$

Solution: The area is base times height. If θ is the angle between a and b, the height over the base a is $\|b\| \sin\theta$, so

$$\text{Area} = \|a\| \, \|b\| \sin\theta \tag{1.1.6.3}$$

Expanding the components in (1.1.6.2) gives $\|a \times b\|^2 = \|a\|^2 \|b\|^2 - \left(a^\top b\right)^2$, and $a^\top b = \|a\| \, \|b\| \cos\theta$, so

$$\begin{aligned}\|a \times b\|^2 &= \|a\|^2 \|b\|^2 \left(1 - \cos^2\theta\right) \\ &= \left(\|a\| \, \|b\| \sin\theta\right)^2\end{aligned} \tag{1.1.6.4}$$

Since $\sin\theta \geq 0$ for $0 \leq \theta \leq \pi$, taking square roots gives (1.1.6.1).

1.1.7 The direction cosines of a line with direction vector d are the components of the unit vector

$$\frac{d}{\|d\|} = \begin{pmatrix} \cos\alpha \\ \cos\beta \\ \cos\gamma \end{pmatrix} \tag{1.1.7.1}$$

1.1.8 A linear fractional (Möbius) function and its matrix are related by

$$f(x) = \frac{ax + b}{cx + d} \longleftrightarrow M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{1.1.8.1}$$

Then $f_M \circ f_N = f_{MN}$ and $f_{kM} = f_M$ for $k \neq 0$ wherever the functions are defined.

Solution: In homogeneous coordinates,

$$\begin{aligned}M \begin{pmatrix} x \\ 1 \end{pmatrix} &= \begin{pmatrix} ax + b \\ cx + d \end{pmatrix} \\ &= (cx + d) \begin{pmatrix} f(x) \\ 1 \end{pmatrix}\end{aligned} \tag{1.1.8.2}$$

so applying f_N followed by f_M multiplies the column vector by N and then by M, up to non-zero scalars, which do not change the ratio of the entries.

1.1.9 If $f \circ f =$ identity on its domain and f maps the domain onto itself, then f is one-one, onto, and

$$f^{-1} = f \tag{1.1.9.1}$$

1.1.10 For angles α, β, multiplying unit complex numbers adds their angles:

$$(\cos\alpha + \mathrm{j}\sin\alpha)(\cos\beta + \mathrm{j}\sin\beta) = \cos(\alpha+\beta) + \mathrm{j}\sin(\alpha+\beta) \tag{1.1.10.1}$$

1.1.11 For $uv < 1$,

$$\tan^{-1} u + \tan^{-1} v = \tan^{-1}\frac{u+v}{1-uv} \tag{1.1.11.1}$$

$$\implies 2\tan^{-1} u = \tan^{-1}\frac{2u}{1-u^2}, \quad |u| < 1 \tag{1.1.11.2}$$

Solution: For $\alpha = \tan^{-1} u$, $\beta = \tan^{-1} v$, $\tan(\alpha+\beta) = \frac{u+v}{1-uv}$. If $uv < 1$, then $\alpha+\beta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, which is the range of $\tan^{-1}$.

1.1.12 For real $u(x) \neq 0$,

$$\frac{d}{dx}\ln|u| = \frac{u'}{u} \tag{1.1.12.1}$$

1.1.13

$$\frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1 \tag{1.1.13.1}$$

1.1.14 If $f'(x) \geq 0$ on an interval and $f'(x) = 0$ only at isolated points, then f is strictly increasing on that interval.

1.1.15 The gradient of $F(x,y)$ at a point P on the curve $F(x,y) = 0$ is normal to the curve. The normal line has direction $\nabla F(P)$:

$$\nabla F = \begin{pmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \end{pmatrix}, \quad x = P + \lambda \nabla F(P) \tag{1.1.15.1}$$

Solution: Let $r(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ trace the curve near P, so $F(x(t), y(t)) = 0$ for all t. Differentiating with the chain rule,

$$\begin{aligned} \frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} &= (\nabla F)^\top \frac{dr}{dt} \\ &= 0 \end{aligned} \tag{1.1.15.2}$$

Since $\frac{dr}{dt}$ is tangent to the curve, $\nabla F(P)$ is perpendicular to the tangent at P, that is, normal to the curve. The line through P with this direction is $x = P + \lambda \nabla F(P)$.

1.1.16 Integration by parts:

$$\int u\,dv = uv - \int v\,du \tag{1.1.16.1}$$

1.1.17 The linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ has the integrating factor $\mu = e^{\int P\,dx}$ and

$$\frac{d}{dx}(\mu y) = \mu Q \tag{1.1.17.1}$$

Solution: Since $\mu = e^{\int P\,dx}$, we have $\mu' = P\mu$. By the product rule,

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu' y + \mu y' \\ &= \mu(y' + Py) \\ &= \mu Q\end{aligned} \tag{1.1.17.2}$$

Integrating both sides gives $\mu y = \int \mu Q\,dx + C$, so

$$y = \frac{1}{\mu}\left(\int \mu Q\,dx + C\right) \tag{1.1.17.3}$$

1.1.18 A separable equation $f(x)\,dx + g(y)\,dy = 0$ is solved by

$$\int f(x)\,dx + \int g(y)\,dy = C \tag{1.1.18.1}$$

1.1.19 The vector β decomposes as $\beta = \beta_1 + \beta_2$ with $\beta_1 \parallel \alpha$ and $\beta_2 \perp \alpha$, where

$$\beta_1 = P\beta, \quad \beta_2 = (I - P)\beta, \quad P = \frac{\alpha\alpha^\top}{\alpha^\top\alpha} \tag{1.1.19.1}$$

Solution: Since $\beta_1 \parallel \alpha$, let $\beta_1 = \lambda\alpha$. Then $\beta_2 = \beta - \lambda\alpha$, and $\beta_2 \perp \alpha$ requires

$$\begin{aligned}\alpha^\top\beta_2 &= \alpha^\top(\beta - \lambda\alpha) \\ &= 0 \\ \implies \lambda &= \frac{\alpha^\top\beta}{\alpha^\top\alpha}\end{aligned} \tag{1.1.19.2}$$

Hence

$$\begin{aligned}\beta_1 &= \frac{\alpha^\top\beta}{\alpha^\top\alpha}\alpha \\ &= \frac{\alpha\alpha^\top}{\alpha^\top\alpha}\beta \\ &= P\beta\end{aligned} \tag{1.1.19.3}$$

$$\begin{aligned}\beta_2 &= \beta - \beta_1 \\ &= (I - P)\beta\end{aligned} \tag{1.1.19.4}$$

Check: $\alpha^\top\beta_2 = \alpha^\top\beta - \frac{\alpha^\top\alpha\,\alpha^\top\beta}{\alpha^\top\alpha} = 0$.

1.1.20 A line with direction d is parallel to the planes $n_1^\top x = c_1$ and $n_2^\top x = c_2$ if

$$\begin{pmatrix} n_1^\top \\ n_2^\top \end{pmatrix} d = 0 \implies d = n_1 \times n_2 \tag{1.1.20.1}$$

The line through P is $x = P + \lambda d$, and its Cartesian form is $\frac{x-p_1}{d_1} = \frac{y-p_2}{d_2} = \frac{z-p_3}{d_3}$.

1.1.21 If X is the number of successes in n independent trials, each with probability p of success and $q = 1 - p$, then

$$\Pr(X = k) = {}^{n}C_k p^k q^{n-k}, \quad k = 0, 1, \ldots, n \tag{1.1.21.1}$$

$$E[X] = \sum_{k=0}^{n} k \Pr(X = k) = np \tag{1.1.21.2}$$

Solution: Since $k\,{}^{n}C_k = n\,{}^{n-1}C_{k-1}$,

$$\begin{aligned} E[X] &= np \sum_{k=1}^{n} {}^{n-1}C_{k-1} p^{k-1} q^{n-k} \\ &= np\,(p+q)^{n-1} \\ &= np \end{aligned} \tag{1.1.21.3}$$

1.1.22 If A and B are invertible, then

$$(AB)^{-1} = B^{-1}A^{-1} \tag{1.1.22.1}$$

Also, $A^{-1} = \frac{\operatorname{adj}(A)}{|A|}$, where $\operatorname{adj}(A)$ is the transpose of the matrix of cofactors.

Solution: $(AB)\left(B^{-1}A^{-1}\right) = A\left(BB^{-1}\right)A^{-1} = I$.

1.1.23 If $V'(h_0) = 0$ and $V''(h_0) < 0$, then V has a local maximum at h_0.

1.1.24 A right circular cone of height h and base radius r is inscribed in a sphere of radius R, with the apex and the base circle on the sphere, then

$$r^2 = h(2R - h), \; V = \frac{\pi}{3} r^2 h = \frac{\pi}{3} h^2 (2R - h), \; 0 < h < 2R \tag{1.1.24.1}$$

1.1.25 The area under $y = f(x) \geq 0$ between $x = a$ and $x = b$ is

$$\int_a^b f(x)\,dx \tag{1.1.25.1}$$

For a circle of radius r, $\int \sqrt{r^2 - x^2}\,dx = \frac{x}{2}\sqrt{r^2 - x^2} + \frac{r^2}{2}\sin^{-1}\frac{x}{r} + c$.

Solution: Divide $[a, b]$ into n strips of width $h = \frac{b-a}{n}$. The strip at x has area approximately $f(x)\,h$, and adding the strips and letting $n \to \infty$ gives the area $\int_a^b f(x)\,dx$, as in (1.1.26.1). For the circle, put $x = r\sin\theta$ with $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so $\sqrt{r^2 - x^2} = r\cos\theta$ and $dx = r\cos\theta\,d\theta$. Then

$$\begin{aligned} \int \sqrt{r^2 - x^2}\,dx &= r^2 \int \cos^2\theta\,d\theta \\ &= \frac{r^2}{2} \int (1 + \cos 2\theta)\;d\theta \\ &= \frac{r^2}{2}(\theta + \sin\theta\cos\theta) + c \end{aligned} \tag{1.1.25.2}$$

Using $\theta = \sin^{-1}\frac{x}{r}$ and $\sin\theta\cos\theta = \frac{x}{r}\cdot\frac{\sqrt{r^2-x^2}}{r}$,

$$\int \sqrt{r^2-x^2}\,dx = \frac{x}{2}\sqrt{r^2-x^2} + \frac{r^2}{2}\sin^{-1}\frac{x}{r} + c \tag{1.1.25.3}$$

1.1.26 A definite integral is the limit of a sum,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} h\sum_{i=1}^{n} f(a+ih), \quad h = \frac{b-a}{n} \tag{1.1.26.1}$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \tag{1.1.26.2}$$

Solution: For (1.1.26.1), divide $[a,b]$ into n equal parts of width $h = \frac{b-a}{n}$ with points $x_i = a+ih$, $i = 0,1,\ldots,n$. On the ith part take a rectangle of height $f(x_i)$; its area is $h\,f(a+ih)$. The total area of the rectangles is

$$S_n = h\sum_{i=1}^{n} f(a+ih) \tag{1.1.26.3}$$

As $n\to\infty$, $h\to 0$, and for a continuous f the sums S_n approach the area under the curve, which is $\int_a^b f(x)\,dx$. For example, for $f(x) = x$ on $[0,1]$, $h = \frac{1}{n}$ and, using (1.1.26.2),

$$\begin{aligned} S_n &= \frac{1}{n}\sum_{i=1}^{n}\frac{i}{n} \\ &= \frac{n+1}{2n} \to \frac{1}{2} = \int_0^1 x\,dx \end{aligned} \tag{1.1.26.4}$$

Solution: For (1.1.26.2), let $S = \sum_{i=1}^{n} i$. Writing the sum forwards and backwards and adding term by term, each of the n pairs adds up to $n+1$, so

$$2S = n(n+1) \implies \sum_{i=1}^{n} i = \frac{n(n+1)}{2} \tag{1.1.26.5}$$

For the second sum, use $(i+1)^3 - i^3 = 3i^2+3i+1$ and add for $i = 1,\ldots,n$. The left side telescopes to $(n+1)^3 - 1$, so

$$(n+1)^3 - 1 = 3\sum_{i=1}^{n} i^2 + 3\frac{n(n+1)}{2} + n \tag{1.1.26.6}$$

Since $(n+1)^3 - 1 - n = n(n+1)(n+2)$,

$$\begin{aligned} 3\sum_{i=1}^{n} i^2 &= n(n+1)(n+2) - \frac{3n(n+1)}{2} \\ &= \frac{n(n+1)(2n+1)}{2} \\ \implies \sum_{i=1}^{n} i^2 &= \frac{n(n+1)(2n+1)}{6} \end{aligned} \tag{1.1.26.7}$$

1.1.27 For $t = \tan x$,

$$dt = \left(1+t^2\right) dx, \quad \frac{\cos^2 x}{\cos^2 x + k\sin^2 x} = \frac{1}{1+kt^2} \tag{1.1.27.1}$$

1.1.28 The plane through the points P_1, P_2 and perpendicular to the plane $n_0^\top x = c_0$ has the normal n with $n^\top (P_2 - P_1) = 0$ and $n^\top n_0 = 0$. Hence

$$n = (P_2 - P_1) \times n_0, \quad n^\top x = n^\top P_1 \tag{1.1.28.1}$$

The plane contains the line $x = A_0 + \lambda d$ if $n^\top A_0 = n^\top P_1$ and $n^\top d = 0$.

1.1.29 (Corner point theorem) In a linear programming problem, the feasible region is convex, and the optimum value of the linear objective $Z = c^\top x$ is attained at a corner point. If the region is unbounded and $Z \geq 0$ on it (for example $c \geq 0$ with $x \geq 0$), the minimum exists and is attained at a corner point.

1.1.30 (Bayes' theorem) For a partition $\{A_1, \ldots, A_n\}$ of the sample space and an event E,

$$\Pr(A_i|E) = \frac{\Pr(A_i)\Pr(E|A_i)}{\sum_{j=1}^{n} \Pr(A_j)\Pr(E|A_j)} \tag{1.1.30.1}$$

Solution: By the definition of conditional probability and the multiplication rule,

$$\Pr(A_i|E) = \frac{\Pr(A_i \cap E)}{\Pr(E)}, \quad \Pr(A_i \cap E) = \Pr(A_i)\Pr(E|A_i) \tag{1.1.30.2}$$

Since $A_1, \ldots, A_n$ are mutually exclusive and exhaustive, E is the disjoint union of the events $A_j \cap E$, so by the total probability rule

$$\begin{aligned} \Pr(E) &= \sum_{j=1}^{n} \Pr(A_j \cap E) \\ &= \sum_{j=1}^{n} \Pr(A_j)\Pr(E|A_j) \end{aligned} \tag{1.1.30.3}$$

Substituting these in the first equation gives (1.1.30.1).

B. Problems

1.2.1 Write the value of $\cot\left(\tan^{-1} a + \cot^{-1} a\right)$.

Solution: From (1.1.1.1),

$$\begin{aligned}\cot\left(\tan^{-1} a + \cot^{-1} a\right) &= \cot\frac{\pi}{2}\\ &= 0\end{aligned} \tag{1.2.1.1}$$

1.2.2 If A is a square matrix such that $A^2 = A$, then write the value of $(I + A)^2 - 3A$.

Solution: From (1.1.2.1) and $A^2 = A$,

$$\begin{aligned}(I + A)^2 - 3A &= I + 2A + A - 3A\\ &= I\end{aligned} \tag{1.2.2.1}$$

The following code verifies the solution using a random idempotent matrix

codes/Q02.py

1.2.3 If $x\begin{pmatrix}2\\3\end{pmatrix} + y\begin{pmatrix}-1\\1\end{pmatrix} = \begin{pmatrix}10\\5\end{pmatrix}$, write the value of x.

Solution: The equation can be expressed as

$$\begin{pmatrix}2 & -1\\3 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}10\\5\end{pmatrix} \tag{1.2.3.1}$$

Reducing the augmented matrix by row operations,

$$\begin{pmatrix}2 & -1 & 10\\3 & 1 & 5\end{pmatrix} \xrightarrow{R_2 \to 2R_2 - 3R_1} \begin{pmatrix}2 & -1 & 10\\0 & 5 & -20\end{pmatrix} \tag{1.2.3.2}$$

Both A and $\begin{pmatrix}A & b\end{pmatrix}$ have two non-zero rows, so $\operatorname{rank} A = \operatorname{rank}\begin{pmatrix}A & b\end{pmatrix} = 2$, which equals the number of unknowns. From (1.1.3.4), the system is consistent and has a unique solution. Back substitution in the reduced matrix gives

$$5y = -20 \implies y = -4 \tag{1.2.3.3}$$

$$2x - y = 10 \implies x = 3 \tag{1.2.3.4}$$

Hence $x = 3$ (and $y = -4$).

The following code verifies the solution

codes/Q03.py

1.2.4 Write the value of the following determinant:

$$\begin{vmatrix}102 & 18 & 36\\1 & 3 & 4\\17 & 3 & 6\end{vmatrix} \tag{1.2.4.1}$$

Solution: Let R_i denote the ith row. Then

$$\begin{aligned} R_1 &= \begin{pmatrix} 102 & 18 & 36 \end{pmatrix} \\ &= 6\begin{pmatrix} 17 & 3 & 6 \end{pmatrix} \\ &= 6R_3 \end{aligned} \tag{1.2.4.2}$$

The row operation $R_1 \to R_1 - 6R_3$ gives a zero row. From (1.1.4.5), the determinant is 0.

1.2.5 If $\int \left(\frac{x-1}{x^2}\right) e^x \, dx = f(x)e^x + c$, then write the value of $f(x)$.

Solution: Write

$$\begin{aligned} \frac{x-1}{x^2} &= \frac{1}{x} - \frac{1}{x^2} \\ &= g(x) + g'(x), \quad g(x) = \frac{1}{x} \end{aligned} \tag{1.2.5.1}$$

From (1.1.5.1),

$$\int \left(\frac{x-1}{x^2}\right) e^x \, dx = \frac{e^x}{x} + c \tag{1.2.5.2}$$

Hence $f(x) = \frac{1}{x}$.

1.2.6 Write the value of the area of the parallelogram determined by the vectors $2\hat{\imath}$ and $3\hat{\jmath}$.

Solution: In components,

$$\begin{aligned} a &= 2\hat{\imath} \\ &= \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \end{aligned} \tag{1.2.6.1}$$

$$\begin{aligned} b &= 3\hat{\jmath} \\ &= \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \end{aligned} \tag{1.2.6.2}$$

From (1.1.6.2),

$$\begin{aligned} a \times b &= \begin{pmatrix} 0 \cdot 0 - 0 \cdot 3 \\ 0 \cdot 0 - 2 \cdot 0 \\ 2 \cdot 3 - 0 \cdot 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} \end{aligned} \tag{1.2.6.3}$$

From (1.1.6.1),

$$\begin{aligned}\text{Area} &= \|a \times b\| \\ &= \left\| \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} \right\| \\ &= 6 \end{aligned} \tag{1.2.6.4}$$

1.2.7 Write the direction cosines of a line parallel to the z-axis.

Solution: A direction vector of the z-axis is

$$d = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \|d\| = 1 \tag{1.2.7.1}$$

From (1.1.7.1), the direction cosines are

$$(\cos\alpha, \cos\beta, \cos\gamma) = (0, 0, 1) \tag{1.2.7.2}$$

(The opposite orientation gives $(0, 0, -1)$.)

1.2.8 If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$, show that $f \circ f(x) = x$ for all $x \neq \frac{2}{3}$. What is the inverse of f?

Solution: From (1.1.8.1), f corresponds to the matrix

$$M = \begin{pmatrix} 4 & 3 \\ 6 & -4 \end{pmatrix} \tag{1.2.8.1}$$

$$\begin{aligned} M^2 &= \begin{pmatrix} 16+18 & 12-12 \\ 24-24 & 18+16 \end{pmatrix} \\ &= 34I \end{aligned} \tag{1.2.8.2}$$

$34I$ corresponds to $\frac{34x}{34} = x$. Hence $f \circ f(x) = x$ for all $x \neq \frac{2}{3}$. Further, $f(x) = \frac{2}{3}$ would require $4x + 3 = 4x - \frac{8}{3}$, which is impossible. So f maps $\mathbb{R} \setminus \left\{\frac{2}{3}\right\}$ into itself. By (1.1.9.1), f is one-one, onto and

$$\begin{aligned} f^{-1}(x) &= f(x) \\ &= \frac{4x+3}{6x-4}, \quad x \neq \frac{2}{3} \end{aligned} \tag{1.2.8.3}$$

1.2.9 Prove that

$$\sin^{-1}\frac{63}{65} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5} \tag{1.2.9.1}$$

Solution: Let $\alpha = \sin^{-1}\frac{5}{13}$ and $\beta = \cos^{-1}\frac{3}{5}$. Then $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, and

$$\cos\alpha = \frac{12}{13}, \quad \sin\beta = \frac{4}{5} \tag{1.2.9.2}$$

From (1.1.10.1),

$$\left(\frac{12+5\mathrm{j}}{13}\right)\left(\frac{3+4\mathrm{j}}{5}\right) = \frac{(36-20)+(48+15)\,\mathrm{j}}{65}$$
$$= \frac{16+63\mathrm{j}}{65} \tag{1.2.9.3}$$

Hence $\cos(\alpha+\beta) = \frac{16}{65} > 0$ and $\sin(\alpha+\beta) = \frac{63}{65}$. Since $\alpha+\beta \in (0,\pi)$ and the cosine is positive, $\alpha+\beta \in \left(0, \frac{\pi}{2}\right)$, which lies in the range of $\sin^{-1}$. Therefore

$$\alpha+\beta = \sin^{-1}\frac{63}{65} \tag{1.2.9.4}$$

OR

Solve for x: $2\tan^{-1}(\sin x) = \tan^{-1}(2\sec x)$, $x \neq \frac{\pi}{2}$.

Solution: The right hand side requires $\cos x \neq 0$, so $|\sin x| < 1$. From (1.1.11.2) with $u = \sin x$,

$$2\tan^{-1}(\sin x) = \tan^{-1}\frac{2\sin x}{1-\sin^2 x}$$
$$= \tan^{-1}\frac{2\sin x}{\cos^2 x} \tag{1.2.9.5}$$

Since $\tan^{-1}$ is one-one,

$$\frac{2\sin x}{\cos^2 x} = \frac{2}{\cos x}$$
$$\implies \sin x = \cos x$$
$$\implies \tan x = 1 \tag{1.2.9.6}$$

Hence $x = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$, and the principal solution is

$$x = \frac{\pi}{4} \tag{1.2.9.7}$$

The following code verifies both parts, including the roots at $\frac{\pi}{4}$ and $\frac{5\pi}{4}$

codes/Q09.py

1.2.10 Using properties of determinants, prove that

$$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix} = a^3 \tag{1.2.10.1}$$

Solution: Let the matrix be M. Apply the row operations $R_2 \to R_2 - 2R_1$ and

$R_3 \to R_3 - 3R_1$,

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix}, \quad E_1 M = \begin{pmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{pmatrix} \tag{1.2.10.2}$$

Then apply $R_3 \to R_3 - 3R_2$,

$$E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix}, \quad E_2 E_1 M = \begin{pmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 0 & a \end{pmatrix} \tag{1.2.10.3}$$

From (1.1.4.1) and (1.1.4.3),

$$\begin{aligned} |M| &= |E_2 E_1 M| \\ &= a \cdot a \cdot a \\ &= a^3 \end{aligned} \tag{1.2.10.4}$$

The following code verifies the solution

codes/Q10.py

1.2.11 If $x^m y^n = (x+y)^{m+n}$, prove that $\frac{dy}{dx} = \frac{y}{x}$.

Solution: Taking logarithms of the absolute values (valid wherever $x, y, x+y \neq 0$)

$$m \ln|x| + n \ln|y| = (m+n) \ln|x+y| \tag{1.2.11.1}$$

Differentiating with respect to x using (1.1.12.1),

$$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{x+y}\left(1 + \frac{dy}{dx}\right) \tag{1.2.11.2}$$

$$\implies \frac{dy}{dx}\left(\frac{n}{y} - \frac{m+n}{x+y}\right) = \frac{m+n}{x+y} - \frac{m}{x} \tag{1.2.11.3}$$

$$\implies \frac{dy}{dx} \cdot \frac{nx - my}{y(x+y)} = \frac{nx - my}{x(x+y)} \tag{1.2.11.4}$$

$$\implies (nx - my)\left(x\frac{dy}{dx} - y\right) = 0 \tag{1.2.11.5}$$

If $nx \neq my$, then $x\frac{dy}{dx} = y$. If $nx = my$ on an interval, then $y = \frac{n}{m}x$ there and $\frac{dy}{dx} = \frac{n}{m} = \frac{y}{x}$. In both cases

$$\frac{dy}{dx} = \frac{y}{x} \tag{1.2.11.6}$$

Remark. For integers m, n, dividing the given relation by x^{m+n} gives $t^n = (1+t)^{m+n}$ with $t = \frac{y}{x}$. This polynomial equation has finitely many roots, so t is constant, $y = tx$ and $\frac{dy}{dx} = t = \frac{y}{x}$. For $m = 2$, $n = 3$, the root $t \approx -2.8906$ is verified by the following code.

codes/Q11.py

1.2.12 If $y = e^{a\cos^{-1}x}$, $-1 \le x \le 1$, show that $\left(1-x^2\right)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$.

Solution: For $-1 < x < 1$, from (1.1.13.1),

$$\frac{dy}{dx} = -\frac{a\,y}{\sqrt{1-x^2}} \implies \sqrt{1-x^2}\,\frac{dy}{dx} = -a\,y \tag{1.2.12.1}$$

Differentiating,

$$\sqrt{1-x^2}\frac{d^2y}{dx^2} - \frac{x}{\sqrt{1-x^2}}\frac{dy}{dx} = -a\frac{dy}{dx} \tag{1.2.12.2}$$

Multiplying by $\sqrt{1-x^2}$ and substituting $\sqrt{1-x^2}\frac{dy}{dx} = -ay$,

$$\begin{aligned}\left(1-x^2\right)\frac{d^2y}{dx^2} - x\frac{dy}{dx} &= -a\left(\sqrt{1-x^2}\frac{dy}{dx}\right)\\ &= a^2y\end{aligned} \tag{1.2.12.3}$$

yielding the required result. (The endpoints $x = \pm 1$ follow by continuity.)

OR

If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, $-1 < x < 1$, $x \neq y$, prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$.

Solution: Writing $x\sqrt{1+y} = -y\sqrt{1+x}$ and squaring,

$$x^2\left(1+y\right) = y^2\left(1+x\right) \implies (x-y)\left(x+y+xy\right) = 0 \tag{1.2.12.4}$$

Since $x \neq y$, $x + y + xy = 0$, so

$$\begin{aligned} y &= -\frac{x}{1+x}\\ \implies \frac{dy}{dx} &= -\frac{(1+x)-x}{(1+x)^2}\\ &= -\frac{1}{(1+x)^2}\end{aligned} \tag{1.2.12.5}$$

This y satisfies the given equation because $1 + y = \frac{1}{1+x}$ gives $x\sqrt{1+y} = \frac{x}{\sqrt{1+x}} = -y\sqrt{1+x}$.

The following code verifies both parts

codes/Q12.py

1.2.13 Show that $y = \log(1+x) - \frac{2x}{2+x}$, $x > -1$, is an increasing function of x throughout its domain.

Solution:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1+x} - \frac{2(2+x)-2x}{(2+x)^2} \\ &= \frac{1}{1+x} - \frac{4}{(2+x)^2} \\ &= \frac{x^2}{(1+x)(2+x)^2}\end{aligned} \tag{1.2.13.1}$$

For $x > -1$ the denominator is positive, so $\frac{dy}{dx} \geq 0$, with equality only at $x = 0$. From (1.1.14), y is strictly increasing on $(-1, \infty)$.

OR

Find the equation of the normal at the point $\left(am^2, am^3\right)$ for the curve $ay^2 = x^3$.

Solution: Let $F(x, y) = x^3 - ay^2$. The point lies on the curve since $a\left(am^3\right)^2 = \left(am^2\right)^3$. From (1.1.15.1),

$$\nabla F = \begin{pmatrix} 3x^2 \\ -2ay \end{pmatrix} \implies \nabla F\left(am^2, am^3\right) = a^2m^3 \begin{pmatrix} 3m \\ -2 \end{pmatrix}, \quad m \neq 0 \tag{1.2.13.2}$$

The normal line through $\left(am^2, am^3\right)$ with direction $\begin{pmatrix} 3m \\ -2 \end{pmatrix}$ is

$$-2\left(x - am^2\right) = 3m\left(y - am^3\right) \implies 2x + 3my - am^2\left(2 + 3m^2\right) = 0 \tag{1.2.13.3}$$

which agrees with the slope $-\frac{2}{3m}$ of the normal, since the tangent slope is $\frac{3m}{2}$.
The following code verifies both parts and generates Fig. 1.2.13.1.

codes/Q13.py

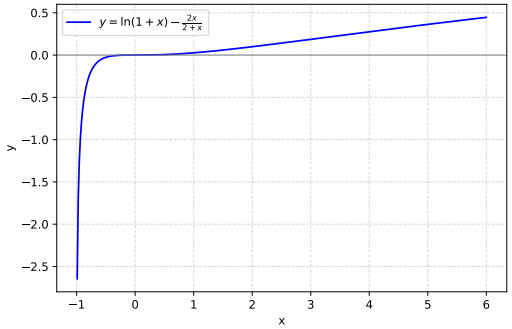


Fig. 1.2.13.1: $y = \ln(1+x) - \frac{2x}{2+x}$ is increasing

1.2.14 Evaluate $\int x^2 \tan^{-1} x \, dx$.

Solution: Using (1.1.16.1) with $u = \tan^{-1} x$, $dv = x^2 \, dx$,

$$\int x^2 \tan^{-1} x \, dx = \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x^3}{1+x^2} \, dx \tag{1.2.14.1}$$

$$\frac{x^3}{1+x^2} = x - \frac{x}{1+x^2} \implies \int \frac{x^3}{1+x^2} \, dx = \frac{x^2}{2} - \frac{1}{2} \ln\left(1+x^2\right) \tag{1.2.14.2}$$

Therefore

$$\int x^2 \tan^{-1} x \, dx = \frac{x^3}{3} \tan^{-1} x - \frac{x^2}{6} + \frac{1}{6} \ln\left(1+x^2\right) + c \tag{1.2.14.3}$$

OR

Evaluate $\int \frac{3x-1}{(x+2)^2} \, dx$.

Solution: Since $3x - 1 = 3\,(x+2) - 7$,

$$\begin{aligned} \int \frac{3x-1}{(x+2)^2} \, dx &= \int \left(\frac{3}{x+2} - \frac{7}{(x+2)^2} \right) dx \\ &= 3 \ln|x+2| + \frac{7}{x+2} + c \end{aligned} \tag{1.2.14.4}$$

The following code verifies both results by differentiation

codes/Q14.py

1.2.15 Solve the differential equation $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \right] \frac{dx}{dy} = 1$, $x \neq 0$.

Solution: The equation requires $x > 0$ and $\frac{dx}{dy} \neq 0$, so it can be written as

$$\frac{dy}{dx} + \frac{1}{\sqrt{x}} y = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} \tag{1.2.15.1}$$

The integrating factor is $\mu = e^{\int x^{-1/2} dx} = e^{2\sqrt{x}}$. From (1.1.17.1),

$$\frac{d}{dx}\left(y e^{2\sqrt{x}} \right) = \frac{1}{\sqrt{x}} \implies y e^{2\sqrt{x}} = 2\sqrt{x} + C \tag{1.2.15.2}$$

Hence

$$y = \left(2\sqrt{x} + C\right) e^{-2\sqrt{x}} \tag{1.2.15.3}$$

The following code verifies the solution

codes/Q15.py

1.2.16 Solve the differential equation $3e^x \tan y\, dx + (2 - e^x) \sec^2 y\, dy = 0$, given that $y = \frac{\pi}{4}$ when $x = 0$.

Solution: Separating the variables,

$$\frac{3e^x}{2 - e^x} dx + \frac{\sec^2 y}{\tan y} dy = 0 \tag{1.2.16.1}$$

From (1.1.18.1),

$$-3 \ln |2 - e^x| + \ln |\tan y| = \ln |C| \implies \tan y = C\,(2 - e^x)^3 \tag{1.2.16.2}$$

At $x = 0$, $y = \frac{\pi}{4}$: $1 = C\,(1)^3$, so $C = 1$. Hence

$$\tan y = (2 - e^x)^3 \implies y = \tan^{-1}\left((2 - e^x)^3\right) \tag{1.2.16.3}$$

which is valid for $e^x < 2$, the interval containing $x = 0$.

The following code compares a numerical solution of the differential equation with this result

codes/Q16.py

1.2.17 If $\alpha = 3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}$ and $\beta = 2\hat{\imath} + \hat{\jmath} - 4\hat{k}$, then express β in the form $\beta = \beta_1 + \beta_2$, where β_1 is parallel to α and β_2 is perpendicular to α.

Solution: Here

$$\alpha = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \quad \beta = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \tag{1.2.17.1}$$

$$\alpha^\top \alpha = 50 \tag{1.2.17.2}$$

$$\begin{aligned} \alpha^\top \beta &= 6 + 4 - 20 \\ &= -10 \end{aligned} \tag{1.2.17.3}$$

From (1.1.19.1),

$$\begin{aligned}\beta_1 &= \frac{\alpha^\top \beta}{\alpha^\top \alpha}\alpha \\ &= -\frac{1}{5}\begin{pmatrix}3\\4\\5\end{pmatrix} \\ &= \begin{pmatrix}-\frac{3}{5}\\-\frac{4}{5}\\-1\end{pmatrix} \qquad (1.2.17.4)\end{aligned}$$

$$\begin{aligned}\beta_2 &= \beta - \beta_1 \\ &= \begin{pmatrix}\frac{13}{5}\\\frac{9}{5}\\-3\end{pmatrix} \qquad (1.2.17.5)\end{aligned}$$

Check: $\alpha^\top \beta_2 = \frac{39+36-75}{5} = 0$. Hence

$$\beta = \left(-\frac{3}{5}\hat{\imath} - \frac{4}{5}\hat{\jmath} - \hat{k}\right) + \left(\frac{13}{5}\hat{\imath} + \frac{9}{5}\hat{\jmath} - 3\hat{k}\right) \qquad (1.2.17.6)$$

The following code verifies the solution

codes/Q17.py

1.2.18 Find the vector and Cartesian equations of the line passing through the point $P = (1, 2, 3)$ and parallel to the planes $r \cdot \left(\hat{\imath} - \hat{\jmath} + 2\hat{k}\right) = 5$ and $r \cdot \left(3\hat{\imath} + \hat{\jmath} + \hat{k}\right) = 6$.

Solution: The normal vectors are

$$n_1 = \begin{pmatrix}1\\-1\\2\end{pmatrix}, \quad n_2 = \begin{pmatrix}3\\1\\1\end{pmatrix} \qquad (1.2.18.1)$$

From (1.1.20.1), d lies in the null space of

$$\begin{pmatrix}1 & -1 & 2\\3 & 1 & 1\end{pmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} \begin{pmatrix}1 & -1 & 2\\0 & 4 & -5\end{pmatrix} \qquad (1.2.18.2)$$

Taking $d_3 = 4$ gives $d_2 = 5$, $d_1 = d_2 - 2d_3 = -3$, which agrees with

$$\begin{aligned}d &= n_1 \times n_2 \\ &= \begin{pmatrix}-3\\5\\4\end{pmatrix} \qquad (1.2.18.3)\end{aligned}$$

The vector equation of the line is

$$r = \left(\hat{\imath} + 2\hat{\jmath} + 3\hat{k}\right) + \lambda\left(-3\hat{\imath} + 5\hat{\jmath} + 4\hat{k}\right) \qquad (1.2.18.4)$$

and the Cartesian equation is

$$\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4} \tag{1.2.18.5}$$

1.2.19 A pair of dice is thrown 4 times. If getting a doublet is considered a success, find the probability distribution of the number of successes and hence find its mean.

Solution: A doublet occurs in 6 of the 36 outcomes, so $p = \frac{1}{6}$ and $q = \frac{5}{6}$. The number of successes X in $n = 4$ independent throws follows (1.1.21.1),

$$\Pr(X = k) = {}^4C_k \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{4-k} \tag{1.2.19.1}$$

k	0	1	2	3	4
$\Pr(X = k)$	$\frac{625}{1296}$	$\frac{500}{1296}$	$\frac{150}{1296}$	$\frac{20}{1296}$	$\frac{1}{1296}$

The probabilities add up to $\frac{1296}{1296} = 1$. From (1.1.21.2),

$$\begin{aligned} E[X] &= np \\ &= 4\left(\frac{1}{6}\right) \\ &= \frac{2}{3} \end{aligned} \tag{1.2.19.2}$$

The following code verifies the solution (including a simulation) and generates Fig. 1.2.19.1.

codes/Q19.py

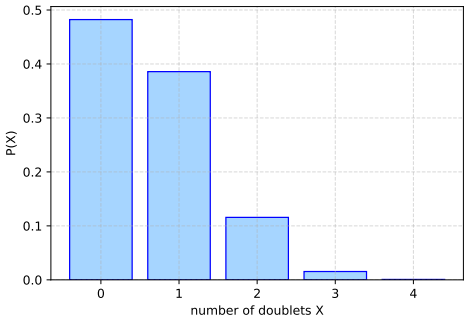


Fig. 1.2.19.1: Probability distribution of the number of doublets

1.2.20 Using matrices, solve the following system of equations: $x-y+z = 4$, $2x+y-3z = 0$, $x+y+z = 2$.

Solution: From (1.1.3.1),

$$\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \tag{1.2.20.1}$$

$$\begin{aligned} \left|A\right| &= 1\left(1+3\right)+1\left(2+3\right)+1\left(2-1\right) \\ &= 10 \neq 0 \end{aligned} \tag{1.2.20.2}$$

so the solution is unique. Gaussian elimination on the augmented matrix gives

$$\begin{pmatrix} 1 & -1 & 1 & 4 \\ 2 & 1 & -3 & 0 \\ 1 & 1 & 1 & 2 \end{pmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 3 & -5 & -8 \\ 0 & 2 & 0 & -2 \end{pmatrix} \tag{1.2.20.3}$$

The last row gives $y = -1$. Then $3\left(-1\right) - 5z = -8 \implies z = 1$ and $x + 1 + 1 = 4 \implies x = 2$. Hence

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \tag{1.2.20.4}$$

OR

If $A^{-1} = \begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{pmatrix}$, find $\left(AB\right)^{-1}$.

Solution: From (1.1.22.1), $\left(AB\right)^{-1} = B^{-1}A^{-1}$. Here $\left|B\right| = 1\left(3\right) - 2\left(-1\right) - 2\left(2\right) = 1$ and, from (1.1.22), the matrix of cofactors is

$$\begin{aligned} C &= \begin{pmatrix} 3 & 1 & 2 \\ 2 & 1 & 2 \\ 6 & 2 & 5 \end{pmatrix} \\ \implies B^{-1} &= C^{\top} \\ &= \begin{pmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{pmatrix} \end{aligned} \tag{1.2.20.5}$$

Therefore

$$\begin{aligned} \left(AB\right)^{-1} &= \begin{pmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \end{aligned} \tag{1.2.20.6}$$

The following code verifies both parts

codes/Q20.py

1.2.21 Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius R is $\frac{4R}{3}$.

Solution: From (1.1.24.1),

$$V(h) = \frac{\pi}{3}\left(2Rh^2 - h^3\right), \quad 0 < h < 2R \tag{1.2.21.1}$$

$$\begin{aligned} V'(h) &= \frac{\pi}{3}\left(4Rh - 3h^2\right) \\ &= \frac{\pi}{3}h\left(4R - 3h\right) \end{aligned} \tag{1.2.21.2}$$

Since $h > 0$, $V'(h) = 0$ gives $h = \frac{4R}{3}$ as the only critical point, and

$$V''(h) = \frac{\pi}{3}(4R - 6h) \implies V''\left(\frac{4R}{3}\right) = -\frac{4\pi R}{3} < 0 \tag{1.2.21.3}$$

From (1.1.23), V is maximum at $h = \frac{4R}{3}$. Since $V \to 0$ as $h \to 0$ or $h \to 2R$, this is the global maximum, with $V_{\max} = \frac{32\pi R^3}{81}$.

The following code verifies the solution and generates Fig. 1.2.21.1.

codes/Q21.py

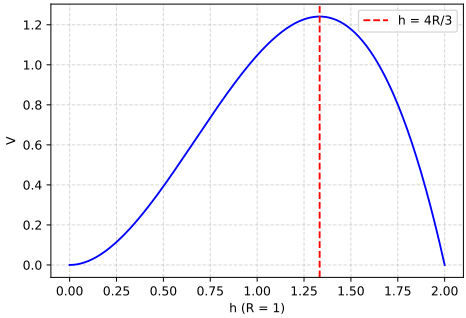


Fig. 1.2.21.1: Volume of the inscribed cone versus its height ($R = 1$)

1.2.22 Find the area of the region in the first quadrant enclosed by the x-axis, the line $x = \sqrt{3}\,y$ and the circle $x^2 + y^2 = 4$.

Solution: The line and the circle meet where $3y^2 + y^2 = 4$, that is, at $\left(\sqrt{3}, 1\right)$. The region is bounded by the line for $0 \le x \le \sqrt{3}$ and by the circle for $\sqrt{3} \le x \le 2$.

From (1.1.25.1),

$$A = \int_0^{\sqrt{3}} \frac{x}{\sqrt{3}}\,dx + \int_{\sqrt{3}}^{2} \sqrt{4-x^2}\,dx \tag{1.2.22.1}$$

$$= \frac{\sqrt{3}}{2} + \left[\frac{x}{2}\sqrt{4-x^2} + 2\sin^{-1}\frac{x}{2}\right]_{\sqrt{3}}^{2} \tag{1.2.22.2}$$

$$= \frac{\sqrt{3}}{2} + \left(\pi - \frac{\sqrt{3}}{2} - \frac{2\pi}{3}\right)$$

$$= \frac{\pi}{3} \tag{1.2.22.3}$$

This agrees with the sector of radius 2 and angle $\frac{\pi}{6}$: $\frac{1}{2}(4)\frac{\pi}{6} = \frac{\pi}{3}$.

The following code verifies the solution and generates Fig. 1.2.22.1.

codes/Q22.py

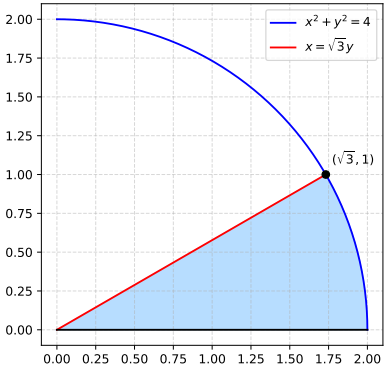


Fig. 1.2.22.1: Region bounded by the x-axis, $x = \sqrt{3}y$ and $x^2 + y^2 = 4$

1.2.23 Evaluate $\int_1^3 \left(x^2 + x\right)\,dx$ as a limit of a sum.

Solution: Here $a = 1$, $b = 3$, $h = \frac{2}{n}$ and $f(1+ih) = (1+ih)^2 + (1+ih) = 2 + 3ih + i^2h^2$. From (1.1.26.1) and (1.1.26.2),

$$S_n = h\left[2n + 3h\frac{n(n+1)}{2} + h^2\frac{n(n+1)(2n+1)}{6}\right] \tag{1.2.23.1}$$

$$= 4 + \frac{6(n+1)}{n} + \frac{4(n+1)(2n+1)}{3n^2}$$

$$= \frac{38}{3} + \frac{10}{n} + \frac{4}{3n^2} \tag{1.2.23.2}$$

Hence

$$\int_1^3 \left(x^2+x\right)\,dx = \lim_{n\to\infty} S_n = \frac{38}{3} \tag{1.2.23.3}$$

OR

Evaluate $\int_0^{\pi/4} \frac{\cos^2 x}{\cos^2 x+4\sin^2 x}\,dx$.

Solution: From (1.1.27.1) with $t=\tan x$ and $k=4$,

$$\int_0^{\pi/4} \frac{\cos^2 x}{\cos^2 x+4\sin^2 x}\,dx = \int_0^1 \frac{dt}{(1+t^2)\,(1+4t^2)} \tag{1.2.23.4}$$

Using partial fractions, $\frac{1}{(1+t^2)(1+4t^2)} = \frac{-1/3}{1+t^2} + \frac{4/3}{1+4t^2}$,

$$\int_0^1 \left(\frac{-1/3}{1+t^2}+\frac{4/3}{1+4t^2}\right)dt = \left[-\frac{1}{3}\tan^{-1}t+\frac{2}{3}\tan^{-1}(2t)\right]_0^1 \tag{1.2.23.5}$$

$$= \frac{2}{3}\tan^{-1}2-\frac{\pi}{12} \tag{1.2.23.6}$$

which is approximately 0.4763. The following code verifies both parts

codes/Q23.py

1.2.24 Find the vector equation of the plane passing through the points $(2,1,-1)$ and $(-1,3,4)$ and perpendicular to the plane $x-2y+4z=10$. Also show that the plane thus obtained contains the line $r=-\hat{\imath}+3\hat{\jmath}+4\hat{k}+\lambda\left(3\hat{\imath}-2\hat{\jmath}-5\hat{k}\right)$.

Solution: Let

$$P_1=\begin{pmatrix}2\\1\\-1\end{pmatrix},\quad P_2=\begin{pmatrix}-1\\3\\4\end{pmatrix},\quad n_0=\begin{pmatrix}1\\-2\\4\end{pmatrix} \tag{1.2.24.1}$$

From (1.1.28.1),

$$\begin{aligned} n &= (P_2 - P_1) \times n_0 \\ &= \begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 8+10 \\ 5+12 \\ 6-2 \end{pmatrix} \\ &= \begin{pmatrix} 18 \\ 17 \\ 4 \end{pmatrix} \end{aligned} \tag{1.2.24.2}$$

$$\begin{aligned} n^\top P_1 &= 36 + 17 - 4 \\ &= 49 \end{aligned} \tag{1.2.24.3}$$

The vector equation of the plane is

$$r \cdot \left(18\hat{\imath} + 17\hat{\jmath} + 4\hat{k}\right) = 49 \tag{1.2.24.4}$$

Check: $n^\top P_2 = -18 + 51 + 16 = 49$. For the given line,

$$A_0 = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}, \quad d = \begin{pmatrix} 3 \\ -2 \\ -5 \end{pmatrix} \tag{1.2.24.5}$$

$$n^\top A_0 = 49 \tag{1.2.24.6}$$

$$\begin{aligned} n^\top d &= 54 - 34 - 20 \\ &= 0 \end{aligned} \tag{1.2.24.7}$$

Hence the line passes through a point of the plane and is parallel to it, so the plane contains the line.

The following code verifies the solution

codes/Q24.py

1.2.25 A company produces soft drinks, and a contract requires that a minimum of 80 units of chemical A and 60 units of chemical B go into each bottle. The chemicals are available in prepared mix packets from two suppliers. A packet from supplier S has 4 units of A and 2 units of B and costs Rs. 10. A packet from supplier T has 1 unit of A and 1 unit of B and costs Rs. 4. How many packets from S and T should the company purchase to honour the contract and yet minimize the cost? Make an LPP and solve it graphically.

Solution: Let $x = \begin{pmatrix} x \\ y \end{pmatrix}$, where x and y are the numbers of packets from S and T.

The LPP is

$$\min_{x} \; Z = \begin{pmatrix} 10 & 4 \end{pmatrix} x \quad \text{s.t.} \quad \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} x \geq \begin{pmatrix} 80 \\ 60 \end{pmatrix}, \quad x \geq 0 \tag{1.2.25.1}$$

The lines $4x + y = 80$ and $2x + y = 60$ meet at $(10, 40)$. The feasible region in Fig. 1.2.25.1 is unbounded, with corner points

Corner point	$(0, 80)$	$(10, 40)$	$(30, 0)$
$Z = 10x + 4y$	320	260	300

Since $Z \geq 0$ on the feasible region, from (1.1.29) the minimum is attained at a corner point. This can also be seen directly:

$$\begin{aligned} 10x + 4y &= (4x + y) + 3\,(2x + y) \\ &\geq 80 + 3\,(60) \\ &= 260 \end{aligned} \tag{1.2.25.2}$$

for every feasible point, with equality at $(10, 40)$. Hence the company should buy 10 packets from S and 40 packets from T, at the minimum cost of Rs. 260.

The following code solves the LPP and generates Fig. 1.2.25.1.

codes/Q25.py

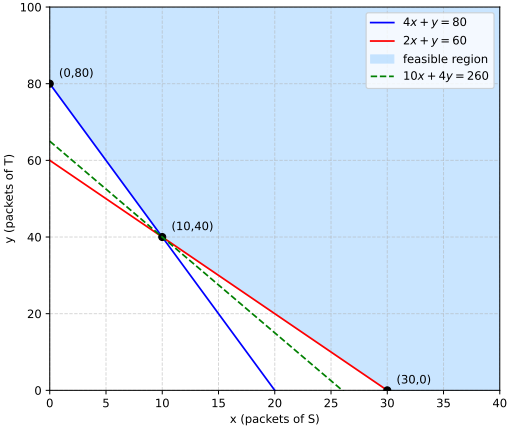


Fig. 1.2.25.1: Feasible region and the optimal corner point $(10, 40)$

1.2.26 In a certain college, 4% of boys and 1% of girls are taller than 1.75 metres. Furthermore, 60% of the students in the college are girls. A student is selected at random from the college and is found to be taller than 1.75 metres. Find the probability that the selected student is a girl.

Solution: Let G: the student is a girl, B: the student is a boy, and T: the student is taller than 1.75 m. Then

$$\Pr(G) = 0.6, \quad \Pr(B) = 0.4, \quad \Pr(T|G) = 0.01, \quad \Pr(T|B) = 0.04 \tag{1.2.26.1}$$

From (1.1.30.1),

$$\begin{aligned} \Pr(G|T) &= \frac{\Pr(G)\Pr(T|G)}{\Pr(G)\Pr(T|G) + \Pr(B)\Pr(T|B)} \\ &= \frac{0.006}{0.006 + 0.016} \\ &= \frac{6}{22} \\ &= \frac{3}{11} \end{aligned} \tag{1.2.26.2}$$